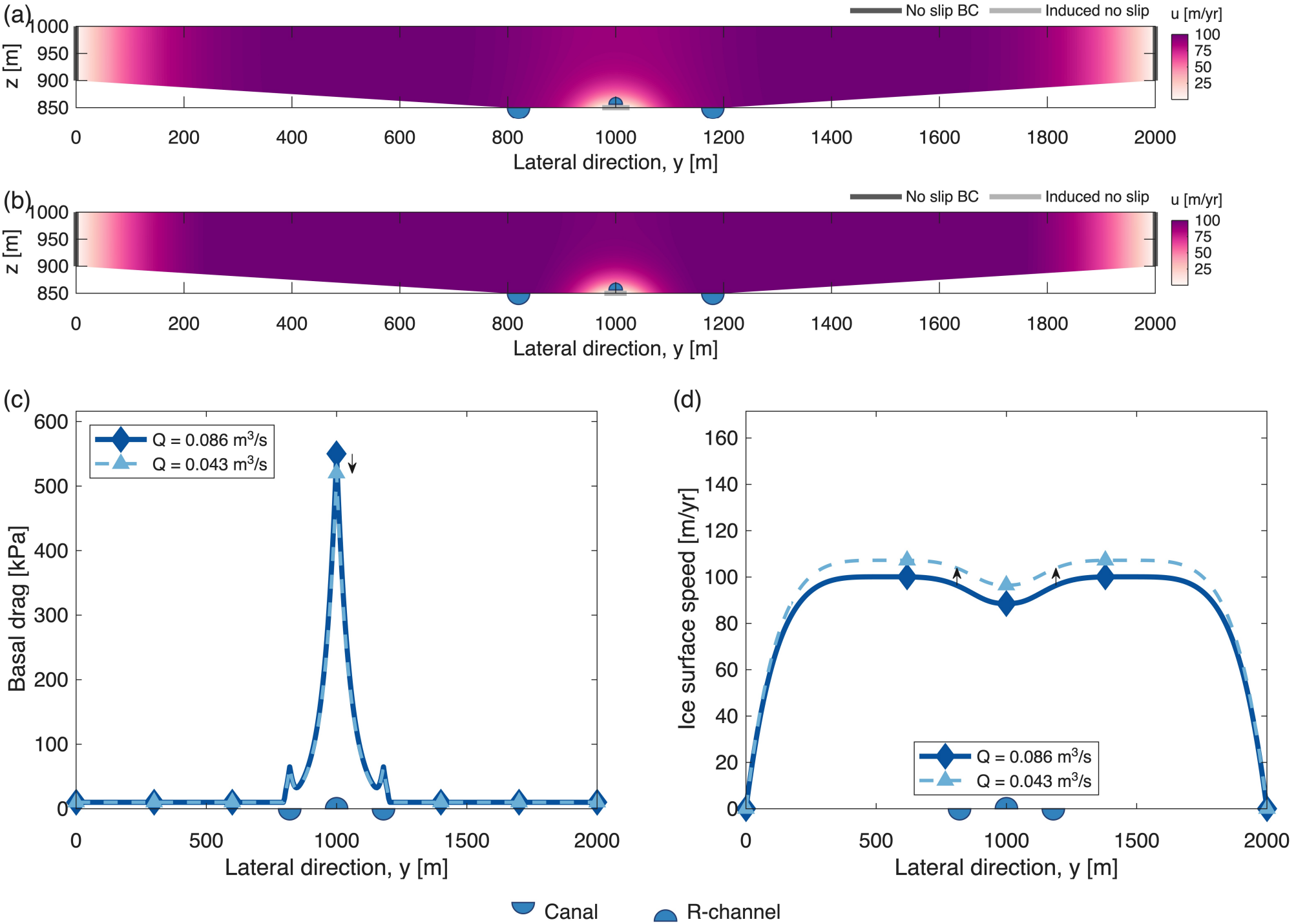

(a)
No slip BC
Induced no slip
u [m/yr]
100
75
50
25
z [m]
1000
950
900
850
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Lateral direction, y [m]
(b)
No slip BC
Induced no slip
u [m/yr]
100
75
50
25
z [m]
1000
950
900
850
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Lateral direction, y [m]
(c)
Q = 0.086 m3/s
Q = 0.043 m3/s
Basal drag [kPa]
600
500
400
300
200
100
0
0
500
1000
1500
2000
Lateral direction, y [m]
(d)
Ice surface speed [m/yr]
160
140
120
100
80
60
40
20
0
Q = 0.086 m3/s
Q = 0.043 m3/s
0
500
1000
1500
2000
Lateral direction, y [m]
Canal
R-channel